

UniHelp daily cost at 3,000 users

3,000 users × 15 queries = 45,000 requests/day. LLM tokens from `benchmarks/fixed.csv` (18 Sep 2026 20:13 IST), not the older `current` harness that still had 8k-token synthesis dumps. Infra is a small always-on app + Postgres, not serverless.

45k

Queries / day

\$4

Likely total / day

\$120

Likely total / month

\$0.00005

LLM cost / query (DeepSeek Flash)

What the fixed run changed

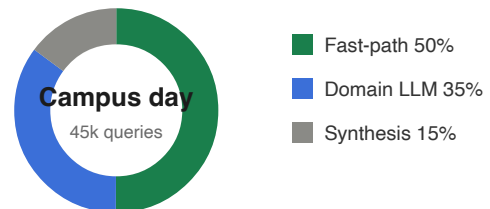
Mess, bus, notices, and timetable singles are all zero tokens (~5 ms). Multi-independent “dinner and bus” replies dropped from 11,058 tokens to 553. Max output in this run is 552 — the ~8,500-token dumps are gone from measured traffic. DeepSeek as-run falls from \$16/day to \$3.59/day, so the harness is no longer an outlier versus a campus day.

Campus-typical mix

50% fast-path lookups (0 tokens), 35% domain-agent LLM (complaint / room / leftover mess), 15% multi-task synthesis. Weighted from `fixed.csv` query shapes, not the 18-prompt harness share.

Share of a campus day, not of the 54-row benchmark. Fast-path is mess, bus, notices, and timetable when the phrasing is obvious. Source:

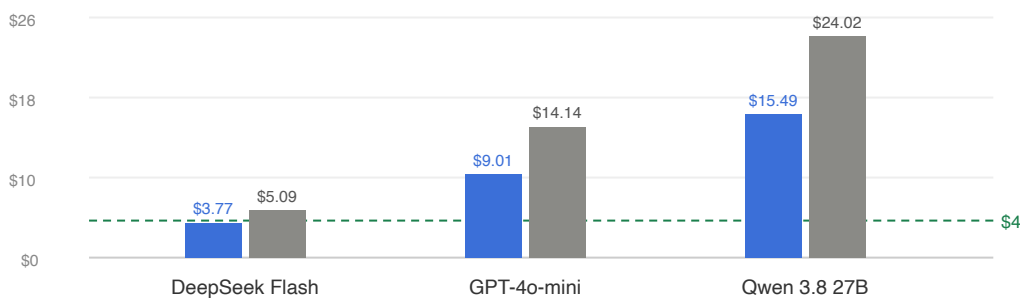
`benchmarks/fixed.csv` + `benchmarks/queries.csv`.



Daily total by model (LLM + \$1.50 infra)

GPT-4o is omitted from the plot — it is ~\$127/day on the same mix and would flatten the useful comparison. The dashed line is the \$4 planning number.

■ Campus-typical (USD/day) ■ Benchmark harness as-run (USD/day) — \$4 plan



DeepSeek V4 Flash 0731 = OpenRouter StreamLake \$0.05412 / \$0.1624 / \$0.001722 per 1M (in / out / cache read), 19 Sep 2026. Qwen 3.8 27B = \$0.15 / \$1.875 / \$0.0375. GPT-4o-mini = \$0.15 / \$0.60 / \$0.075.

GPT-4o on every request is ~\$127/day

Same campus mix, OpenAI list \$2.50 / \$10.00 / \$1.25 per 1M. That is ~\$3,800/month before infra. Do not put a frontier model on mess and bus lookups.

LLM only — four traffic mixes

Mix	DeepSeek Flash	GPT-4o-mini	Qwen 3.8 27B	GPT-4o
Campus-typical (use this)	\$2.27 · ₹200	\$7.51 · ₹661	\$14 · ₹1,231	\$125 · ₹11,017
70% fast-path campus day	\$1.32 · ₹116	\$4.40 · ₹387	\$8.27 · ₹728	\$73 · ₹6,451
Benchmark harness as-run	\$3.59 · ₹316	\$13 · ₹1,112	\$23 · ₹1,982	\$211 · ₹18,543
Campus + 8% 8k-token synthesis (not observed)	\$7.24 · ₹637	\$26 · ₹2,277	\$71 · ₹6,281	\$431 · ₹37,945

USD/day and ≈ INR/day at ₹88/USD. Infra is not in this table. `fixed.csv` has zero rows with output > 552 tokens, so the last row is a residual risk (executor synthesis is still uncapped), not the live measurement.

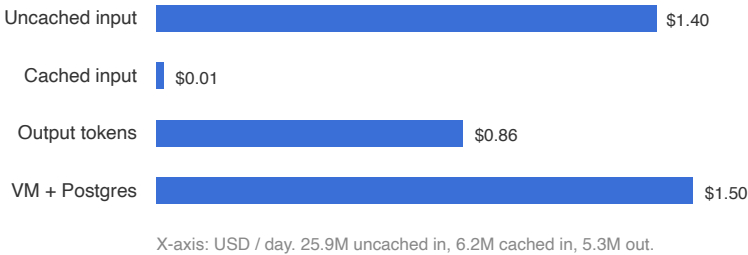
DeepSeek Flash — before vs after the dump fix

Mix	current.csv (before)	fixed.csv (now)
Campus-typical LLM / day	\$2.41	\$2.27
Harness as-run LLM / day	\$16.09	\$3.59
Campus + 8% 8k dumps	\$7.37 (measured mix)	\$7.24 (hypothetical)

Mean tokens/request 3,832 → 1,547 (–60%). Multi-independent 11,058 → 553 (–95%). Single-query LLM calls 0.39 → 0.00 (notices joined the fast path). Source: `benchmarks/comparison_fixed.md`.

Where a \$4 day goes

Campus-typical + DeepSeek Flash 0731 + \$1.50 infra. Almost all of the LLM line is uncached input; prompt cache is already cheap, and output is no longer the dump-driven half of the bill.



X-axis: USD / day. 25.9M uncached in, 6.2M cached in, 5.3M out.

Cost by query shape · per request

Shape	DeepSeek / q	If all 45k
Mess / bus / notices / timetable	\$0	\$0
Multi synthesis (dinner + bus)	\$0.00004	\$1.98
Domain agent (complaint / room)	\$0.00013	\$5.87
Ambiguous / multi-round	\$0.00019	\$8.34

The old “multi + 8k synthesis” shape was ~\$0.0014 / q (\$63 if all 45k). It does not appear in `fixed.csv`.

Setup comparison

Setup	LLM / day	Total / day (incl. ~\$1.50 infra)
Kado gateway free / subsidized	\$0	\$1–2.50
Both tiers DeepSeek Flash (recommended)	\$2.27	~\$4
Live .env: Qwen planner + Flash answers	~\$3–4	~\$5
GPT-4o-mini everywhere	\$7.51	~\$9
GPT-4o on every request	\$125	~\$127

`config.py` and `.env.example` default to OpenRouter `deepseek/deepseek-v4-flash-0731`. This machine's `.env` still sets `LLM_FAST_MODEL=qwen/qwen3.8-27b`. Planner-only Qwen, capped at `FAST_TIER_MAX_TOKENS=800`, adds about \$1/day on the harness and less on a campus mix. Putting Qwen on user-facing answers is the expensive path (\$14 LLM / day campus-typical).

Infra (almost independent of query count at this scale)

45k requests/day is 0.5 QPS average. Meal-time bursts are mostly fast-path DB reads. The app already waits on an external LLM, so you want an always-on process, not Vercel-style billed duration.

Line	What	Daily	Monthly
App server	2 vCPU / 4–8 GB always-on (Hetzner / Railway / a small cloud VM)	\$0.40–1.20	\$12–35
Postgres	Eight small DBs, ~3k users. Same VM or Neon/RDS tiny	\$0.30–0.80	\$10–25
Backups / egress	Campus payload is tiny	\$0.10–0.30	\$3–10
● Infra total	Floor if the LLM line is \$0	\$1.00–2.50	\$30–75

Monthly (30 days)

Kado free + infra only

\$30–75

₹2,640–₹6,600. LLM billed at \$0.

DeepSeek Flash, campus mix

~\$120

₹5,993 LLM + ₹3,960 infra. Planning number.

GPT-4o, same mix

~\$3.8k

Do not put a frontier model on every campus lookup.

Assumptions that move the bill

Fast-path now zeros mess, bus, notices, and timetable when the phrasing is obvious. If real traffic is 70%+ those lookups, the DeepSeek LLM line drops toward \$1.30/day.

Paid estimates use OpenRouter DeepSeek Flash 0731. There is still no public per-token list for `LLM_MODEL=kado` on `awesome.kado.so`; if that gateway is free, you only pay infra.

`LLM_FAST_MODEL=qwen/qwen3.8-27b` is not a cheaper planner. Qwen output is ~11× DeepSeek (\$1.875 vs \$0.162 / 1M). Keep both tiers on Flash unless you need Qwen for quality. The 800-token fast-tier cap stops planner dumps; it does not cap executor synthesis.

Cap completion tokens on executor synthesis if the 8k-token bug class returns. That single failure mode still triples a campus-typical DeepSeek day if ~8% of live traffic hits it.

FX ₹88/USD is a round conversion, not a live rate. Prices are list, not committed discounts.

Generated from the UniHelp daily cost canvas · tokens: benchmarks/fixed.csv · prices: OpenRouter / OpenAI list, 19 Sep 2026.